

PAST EXAMINATION QUESTIONS (FP2)

TOPIC 17: COMPLEX NUMBERS

1. (a) Use de Moivre's theorem to show that $\sin^4 \theta = \frac{1}{8}(\cos 4\theta - 4 \cos 2\theta + 3)$. [5]

Oct/Nov 2020/21/23/6 (part)

2. (a) Show that

$$\sum_{r=1}^n z^{2r} = \frac{z^{2n+1} - z}{z - z^{-1}},$$

for $z \neq 0, 1, -1$. [2]

- (b) By letting $z = \cos \theta + i \sin \theta$, show that, if $\sin \theta \neq 0$,

$$1 + 2 \sum_{r=1}^n \cos(2r\theta) = \frac{\sin(2n+1)\theta}{\sin \theta}. \quad [5]$$

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3. (a) Find the roots of the equation $z^3 = -1 - i$, giving your answers in the form $re^{i\theta}$, where $r > 0$ and $0 \leq \theta < 2\pi$. $\left(z_1 = 2^{\frac{1}{6}}e^{i\frac{5}{12}\pi}, z_2 = 2^{\frac{1}{6}}e^{i\frac{13}{12}\pi}, z_3 = 2^{\frac{1}{6}}e^{i\frac{21}{12}\pi}\right)$ [5]

Let $w = z_1^{3k} + z_2^{3k} + z_3^{3k}$, where k is a positive integer and z_1, z_2, z_3 are roots of $z^3 = -1 - i$.

- (b) Express w in the form $Re^{i\alpha}$, where $R > 0$, giving R and α in terms of k . $\left(R = |w| = 3\left(2^{\frac{1}{2}k}\right), \alpha = \frac{5}{4}k\pi\right)$ [3]

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4. (a) Use de Moivre's theorem to show that $\sin^6 \theta = -\frac{1}{32}(\cos 6\theta - 6 \cos 4\theta + 15 \cos 2\theta - 10)$. [6]

It is given that $\cos^6 \theta = \frac{1}{32}(\cos 6\theta + 6 \cos 4\theta + 15 \cos 2\theta + 10)$.

- (b) Find the exact value of $\int_0^{\frac{1}{3}\pi} \left(\cos^6\left(\frac{1}{4}x\right) + \sin^6\left(\frac{1}{4}x\right)\right) dx$. $\left(\frac{1}{8}\left(\frac{5}{3}\pi + \frac{3}{2}\sqrt{3}\right)\right)$ [4]

- (c) Express each root of the equation $16c^6 + 16(1 - c^2)^3 - 13 = 0$ in the form $\cos k\pi$, where k is a rational number. $\left(\cos\left(\frac{1}{12}\pi\right), \cos\left(\frac{5}{12}\pi\right), \cos\left(\frac{7}{12}\pi\right), \cos\left(\frac{11}{12}\pi\right)\right)$ [5]

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5. (i) Use de Moivre's theorem to show that

$$\sec 6\theta = \frac{\sec^6 \theta}{32 - 48 \sec^2 \theta + 18 \sec^4 \theta - \sec^6 \theta}. \quad [6]$$

- (ii) Hence obtain the roots of the equation

$$3x^6 - 36x^4 + 96x^2 - 64 = 0$$

in the form $\sec q\pi$, where q is rational. $\left(x = \sec q\pi, q = \frac{5}{18}, \frac{7}{18}, \frac{11}{18}, \frac{13}{18}, \frac{17}{18}\right)$ [5]

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6. (i) *****Remove from 9231/2 A2 syllabus*****

Prove by mathematical induction that, for $z \neq 1$ and all positive integers n ,

$$1 + z + z^2 + \dots + z^{n-1} = \frac{z^n - 1}{z - 1}. \quad [5]$$

(ii) By letting $z = \frac{1}{2}(\cos \theta + i \sin \theta)$, use the de Moivre's theorem to deduce that

$$\sum_{m=1}^{\infty} \left(\frac{1}{2}\right)^m \sin m\theta = \frac{2 \sin \theta}{5 - 4 \cos \theta}. \quad [5]$$

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7. (i) Write down the fifth roots of unity. $\left(\exp\left(i \frac{2k\pi}{5}\right), k = 0, \pm 1, \pm 2\right)$ [2]

(ii) Find all the roots of the equation

$$z^{10} + z^5 + 1 = 0,$$

giving each root in the form $e^{i\theta}$. [5]

$$\left(\exp\left(\pm i \frac{2\pi}{15}\right), \exp\left(\pm i \frac{4\pi}{15}\right), \exp\left(\pm i \frac{8\pi}{15}\right), \exp\left(\pm i \frac{2\pi}{3}\right), \exp\left(\pm i \frac{14\pi}{15}\right)\right)$$

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8. (i) Use de Moivre's theorem to show that

$$\sin 8\theta = 8 \sin \theta \cos \theta (1 - 10 \sin^2 \theta + 24 \sin^4 \theta - 16 \sin^6 \theta). \quad [6]$$

(ii) Use the equation $\frac{\sin 8\theta}{\sin 2\theta} = 0$ to find the roots of

$$16x^4 - 24x^4 + 10x^2 - 1 = 0$$

in the form $\sin k\pi$, where k is rational. $\left(k = \frac{1}{8}, \frac{2}{8}, \frac{3}{8}, \frac{9}{8}, \frac{10}{8}, \frac{11}{8}\right)$ [4]

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9. (i) By considering the binomial expansion of $\left(z + \frac{1}{z}\right)^6$, where $z = \cos \theta + i \sin \theta$, express $\cos^6 \theta$ in the form

$$\frac{1}{32}(p + q \cos 2\theta + r \cos 4\theta + s \cos 6\theta),$$

where p, q, r and s are integers to be determined. $(p = 10, q = 15, r = 6, s = 1)$ [6]

(ii) Hence find the exact value of

$$\int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \cos^6\left(\frac{1}{2}x\right) dx. \quad [4]$$

$$\left(\frac{1}{16}\left(5\pi + \frac{44}{3}\right)\right)$$

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10. (i) Show that if $z = e^{i\theta}$, and $z \neq -1$, then

$$\frac{z-1}{z+1} = i \tan \frac{1}{2}\theta. \quad [3]$$

- (ii) Hence, or otherwise, show that if z is a cube root of unity then

$$\frac{z^3-1}{z^3+1} + \frac{z^2-1}{z^2+1} + \frac{z-1}{z+1} = 0. \quad [5]$$

- (iii) Hence write down three roots of the equation

$$(z^3-1)(z^2+1)(z+1) + (z^2-1)(z^3+1)(z+1) + (z-1)(z^3+1)(z^2+1) = 0$$

and find the other three roots. Give your answers in an exact form. $\left(z = 1, e^{\pm \frac{2\pi}{3}i}, z = -1, \frac{1}{3} \pm \frac{2\sqrt{2}}{3}i\right)$ [6]

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11. (i) Use de Moivre's theorem to show that

$$\cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta. \quad [3]$$

- (ii) Hence find all the roots of the equation

$$x^4 - 6x^2 + 1 = 0$$

in the form $\tan q\pi$, where q is a positive rational number. [2]

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12. (i) Use de Moivre's theorem to show that

$$\sin 5\theta = 5 \sin \theta - 20 \sin^3 \theta + 16 \sin^5 \theta. \quad [5]$$

- (ii) Hence explain why the roots of the equation $16x^4 - 20x^2 + 5 = 0$ are $x = \pm \sin \frac{1}{5}\pi$ and $x = \pm \sin \frac{2}{5}\pi$. [3]

- (iii) Without using a calculator, find the exact values of

$$\sin \frac{1}{5}\pi \sin \frac{2}{5}\pi \sin \frac{3}{5}\pi \sin \frac{4}{5}\pi \quad \text{and} \quad \sin^2\left(\frac{1}{5}\pi\right) + \sin^2\left(\frac{2}{5}\pi\right). \quad [4]$$

$$\left(\frac{5}{16}, \frac{5}{4}\right)$$

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13. (i) Let $z = \cos \theta + i \sin \theta$. Show that $z - \frac{1}{z} = 2i \sin \theta$ and hence express $16 \sin^5 \theta$ in the form $\sin 5\theta + p \sin 3\theta + q \sin \theta$, where p and q are integers to be determined. ($p = -5, q = 10$) [6]

- (ii) Hence find the exact value of $\int_0^{\frac{1}{3}\pi} 16 \sin^5 \theta d\theta$. ($\frac{53}{30}$) [3]

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14. (i) Use de Moivre's theorem to prove that

$$\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}. \quad [5]$$

- (ii) Hence find the solutions of the equation $t^4 - 4t^3 - 6t^2 + 4t + 1 = 0$, giving your answers in the form $\tan k\pi$, where k is a rational number. ($k = \frac{3}{16}, \frac{7}{16}, \frac{11}{16}, \frac{15}{16}$) [5]

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15. Let $z = \cos \theta + i \sin \theta$. Show that

$$z^n + \frac{1}{z^n} = 2 \cos n\theta \quad \text{and} \quad z^n - \frac{1}{z^n} = 2i \sin n\theta. \quad [2]$$

By considering $\left(z - \frac{1}{z}\right)^4 \left(z + \frac{1}{z}\right)^2$, show that

$$\sin^4 \theta \cos^2 \theta = \frac{1}{32} (\cos 6\theta - 2 \cos 4\theta - \cos 2\theta + 2). \quad [7]$$

- Hence find the exact value of $\int_0^{\frac{1}{4}\pi} \sin^4 \theta \cos^2 \theta d\theta$. ($\frac{3\pi-4}{192}$) [3]

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16. Use de Moivre's theorem to express $\cot 7\theta$ in terms of $\cot \theta$.

Use the equation $\cot 7\theta = 0$ to show that the roots of the equation

$$x^6 - 21x^4 + 35x^2 - 7 = 0$$

are $\cot\left(\frac{1}{14}k\pi\right)$ for $k = 1, 3, 5, 9, 11, 13$, and deduce that

$$\cot^2\left(\frac{1}{14}\pi\right) \cot^2\left(\frac{3}{14}\pi\right) \cot^2\left(\frac{5}{14}\pi\right) = 7.$$

$$\left(\cot 7\theta = \frac{\cot^7 \theta - 21 \cot^5 \theta + 35 \cot^3 \theta - 7 \cot \theta}{7 \cot^6 \theta - 35 \cot^4 \theta + 21 \cot^2 \theta - 1}\right)$$

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17. Use de Moivre's theorem to show that $\cos^4 \theta = \frac{1}{8} (\cos 4\theta + 4 \cos 2\theta + 3)$. [4]

Find the corresponding expression for $\sin^4 \theta$ in terms of $\cos 4\theta$ and $\cos 2\theta$. $\left(\sin^4 \theta = \frac{1}{8}(\cos 4\theta - 4 \cos 2\theta + 3) \right)$ [4]

Hence find the exact value of $\int_0^{\frac{1}{8}\pi} (\cos^4 \theta + \sin^4 \theta) d\theta$. $\left(\frac{1}{16} + \frac{3}{32} \pi \right)$ [3]

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18. Using de Moivre's theorem, show that

$$\tan 5\theta = \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}. \quad [5]$$

Hence show that the equation $x^2 - 10x + 5 = 0$ has roots $\tan^2 \left(\frac{1}{5} \pi \right)$ and $\tan^2 \left(\frac{2}{5} \pi \right)$. [4]

Deduce a quadratic equation, with integer coefficients, having roots $\sec^2 \left(\frac{1}{5} \pi \right)$ and $\sec^2 \left(\frac{2}{5} \pi \right)$. $(y^2 - 12y + 16 = 0)$ [3]

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19. By considering $\sum_{r=1}^n z^{2r-1}$, where $z = \cos \theta + i \sin \theta$, show that, if $\sin \theta \neq 0$,

$$\sum_{r=1}^n \sin(2r-1)\theta = \frac{\sin^2 n\theta}{\sin \theta}. \quad [7]$$

Deduce that

$$\sum_{r=1}^n (2r-1) \cos \left[\frac{(2r-1)\pi}{2n} \right] = -\operatorname{cosec} \left(\frac{\pi}{2n} \right) \cot \left(\frac{\pi}{2n} \right). \quad [4]$$

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20. Let $z = \cos \theta + i \sin \theta$. Use the binomial expansion of $(1+z)^n$, where n is a positive integer, to show that

$$\binom{n}{1} \cos \theta + \binom{n}{2} \cos 2\theta + \cdots + \binom{n}{n} \cos n\theta = 2^n \cos^n \left(\frac{1}{2} \theta \right) \cos \left(\frac{1}{2} n\theta \right) - 1. \quad [7]$$

Find

$$\binom{n}{1} \sin \theta + \binom{n}{2} \sin 2\theta + \cdots + \binom{n}{n} \sin n\theta. \quad [2]$$

$$\left(2^n \cos^n \frac{\theta}{2} \sin \frac{n\theta}{2} \right)$$

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21. Use de Moivre's theorem to show that

$$[\cos 5\theta \equiv \cos \theta (16 \sin^4 \theta - 12 \sin^2 \theta + 1)]. \quad [5]$$

By considering the equation $\cos 5\theta = 0$, show that the exact value of $\sin^2 \left(\frac{1}{10} \pi \right)$ is $\frac{3-\sqrt{5}}{8}$. [4]

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22. Use de Moivre's theorem to show that

$$\tan 5\theta = \frac{5t - 10t^3 + t^5}{1 - 10t^2 + 5t^4}, \quad [4]$$

where $t = \tan \theta$.

Deduce that the roots of the equation $t^4 - 10t^2 + 5 = 0$ are $\pm \tan \frac{1}{5} \pi$ and $\pm \tan \frac{2}{5} \pi$. [3]

Hence show that $\tan \frac{1}{5} \pi \tan \frac{2}{5} \pi = \sqrt{5}$. [2]

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23. State the sum of the series $z + z^2 + z^3 + \dots + z^n$, for $z \neq 1$. $\left(\frac{z(z^n-1)}{z-1} \right)$ [1]

By letting $z = \cos \theta + i \sin \theta$, show that

$$\cos \theta + \cos 2\theta + \cos 3\theta + \dots + \cos n\theta = \frac{\sin \frac{1}{2} n\theta}{\sin \frac{1}{2} \theta} \cos \frac{1}{2} (n+1)\theta,$$

where $\sin \frac{1}{2} \theta \neq 0$. [7]

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24. State the fifth roots of unity in the form $\cos \theta + i \sin \theta$, where $-\pi < \theta \leq \pi$. $\left(z = \cos \frac{2k\pi}{5} + i \sin \frac{2k\pi}{5}, \text{ where } k = 0, \pm 1, \pm 2 \right)$ [2]

Simplify

$$\left(x - \left[\cos \frac{2}{5} \pi + i \sin \frac{2}{5} \pi \right] \right) \left(x - \left[\cos \frac{2}{5} \pi - i \sin \frac{2}{5} \pi \right] \right). \quad [2]$$

$$\left(x^2 - 2 \cos \frac{2\pi}{5} x + 1 \right)$$

Hence find the real factors of

$$x^5 - 1. \quad [2]$$

$$\left(x^2 - 2 \cos \frac{2\pi}{5} x + 1 \right) \left(x^2 - 2 \cos \frac{4\pi}{5} x + 1 \right) (x - 1)$$

Express the six roots of the equation

$$x^6 - x^3 + 1 = 0$$

as three conjugate pairs, in the form $\cos \theta \pm i \sin \theta$. [6]

$$\left(x = \cos \frac{\pi}{9} \pm i \sin \frac{\pi}{9}; \cos \frac{7\pi}{9} \pm i \sin \frac{7\pi}{9}; \cos \frac{13\pi}{9} \pm i \sin \frac{13\pi}{9} \right)$$

Hence find the real factors of

$$x^6 - x^3 + 1. [2]$$

$$\left(x^2 - 2 \cos \frac{\pi}{9} x + 1 \right) \left(x^2 - 2 \cos \frac{7\pi}{9} x + 1 \right) \left(x^2 - 2 \cos \frac{13\pi}{9} x + 1 \right)$$

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25. Prove by mathematical induction that, for every positive integers n ,

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta. [5]$$

Express $\sin^5 \theta$ in the form $p \sin 5\theta + q \sin 3\theta + r \sin \theta$, where p, q and r are rational numbers to be determined. $\left(p = \frac{1}{16}, q = -\frac{5}{16}, r = \frac{10}{16} \right)$ [6]

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26. By considering the binomial expansion $\left(z - \frac{1}{z} \right)^6$, where $z = \cos \theta + i \sin \theta$, express $\sin^6 \theta$ in the form

$$\frac{1}{32}(p + q \cos 2\theta + r \cos 4\theta + s \cos 6\theta), [6]$$

where p, q, r and s are integers to be determined. $(p = 10, q = -15, r = 6, s = -1)$

Hence find the exact value of $\int_0^{\frac{1}{4}\pi} \sin^6 \theta d\theta$. $\left(\frac{5\pi}{64} - \frac{11}{48} \right)$ [4]

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27. Show the cube root of 1 on an Argand diagram. [1]

Show that the two non-real roots can be expressed in the form ω and ω^2 , and find those cube roots in exact cartesian form $x + iy$. $\left(\omega = -\frac{1}{2} + i\frac{\sqrt{3}}{2}, \omega^2 = -\frac{1}{2} - i\frac{\sqrt{3}}{2} \right)$ [3]

Evaluate the determinant

$$\begin{vmatrix} 1 & 3\omega & 2\omega^2 \\ 3\omega^2 & 2 & \omega \\ 2\omega & \omega^2 & 3 \end{vmatrix}. [3]$$

(-18)

It is given that $z = (4\sqrt{3})\left(\cos\frac{4}{3}\pi + i\sin\frac{4}{3}\pi\right) - 4\left(\cos\frac{11}{6}\pi + i\sin\frac{11}{6}\pi\right)$. Express z in the form $r(\cos\theta + i\sin\theta)$, giving the exact values for r and θ . $\left(z = 8\left(\cos\frac{7}{6}\pi + i\sin\frac{7}{6}\pi\right)\right)$ [5]

Hence find the cube roots of z in the form $r(\cos\theta + i\sin\theta)$. $\left(2\left(\cos\frac{7}{18}\pi + i\sin\frac{7}{18}\pi\right), 2\left(\cos\frac{19}{18}\pi + i\sin\frac{19}{18}\pi\right), 2\left(\cos\frac{31}{18}\pi + i\sin\frac{31}{18}\pi\right)\right)$ [2]

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28. Use de Moivre's theorem to show that

$$\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1. \quad [3]$$

Without using a calculator, verify that $\cos 4\theta = -\cos 3\theta$ for each of the values $\theta = \frac{1}{7}\pi, \frac{3}{7}\pi, \frac{5}{7}\pi, \pi$. [2]

Using the result $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$, show that the roots of the equation

$$8c^4 + 4c^3 - 8c^2 - 3c + 1 = 0$$

are $\cos\frac{1}{7}\pi, \cos\frac{3}{7}\pi, \cos\frac{5}{7}\pi, -1$. [2]

Deduce that $\cos\frac{1}{7}\pi + \cos\frac{3}{7}\pi + \cos\frac{5}{7}\pi = \frac{1}{2}$. [2]

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29. Let $z = \cos\theta + i\sin\theta$. Show that

$$1 + z = 2 \cos\frac{1}{2}\theta \left(\cos\frac{1}{2}\theta + i\sin\frac{1}{2}\theta\right). \quad [3]$$

By considering $(1 + z)^n$, where n is a positive integer, deduce the sum of the series

$$\binom{n}{1} \sin\theta + \binom{n}{2} \sin 2\theta + \dots + \binom{n}{n} \sin n\theta. \quad [6]$$

$$\left(2^n \cos^n \frac{1}{2}\theta \sin \frac{n}{2}\theta\right)$$

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30. Write down the values of θ , in the interval $0 \leq \theta < 2\pi$, for which $\cos\theta + i\sin\theta$ is a fifth root of unity. $\left(\theta = \frac{2k\pi}{5}, k = 0, 1, 2, 3, 4\right)$ [2]

By writing the equation $(z + 1)^5 = z^5$ in the form

$$\left(\frac{z+1}{z}\right)^5 = 1,$$

show that its roots are

$$-\frac{1}{2}\left\{1 + i \cot\left(\frac{k\pi}{5}\right)\right\}, \quad k = 1, 2, 3, 4. \quad [7]$$

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- 31.** Expand $\left(z + \frac{1}{z}\right)^4 \left(z - \frac{1}{z}\right)^2$ and, by substituting $z = \cos \theta + i \sin \theta$, find integers p, q, r, s such that

$$64 \sin^2 \theta \cos^4 \theta = p + q \cos 2\theta + r \cos 4\theta + s \cos 6\theta. \quad [6]$$

$$\left(\left(z^6 + \frac{1}{z^6}\right) + 2\left(z^4 + \frac{1}{z^4}\right) - \left(z^2 + \frac{1}{z^2}\right) - 4; p = 4, q = 2, r = -4, s = -2\right)$$

Using the substitution $x = 2 \cos \theta$, show that

$$\int_1^2 x^4 \sqrt{4 - x^2} dx = \frac{4}{3} \pi + \sqrt{3}. \quad [4]$$

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- 32.** Let $\omega = \cos \frac{1}{5} \pi + i \sin \frac{1}{5} \pi$. Show that $\omega^5 + 1 = 0$ and deduce that

$$\omega^4 - \omega^3 + \omega^2 - \omega = -1. \quad [2]$$

Show further that

$$\omega - \omega^4 = 2 \cos \frac{1}{5} \pi \quad \text{and} \quad \omega^3 - \omega^2 = 2 \cos \frac{3}{5} \pi. \quad [4]$$

Hence find the values of

$$\cos \frac{1}{5} \pi + \cos \frac{3}{5} \pi \quad \text{and} \quad \cos \frac{1}{5} \pi \cos \frac{3}{5} \pi. \quad [4]$$

$$\left(\frac{1}{2}, -\frac{1}{4}\right)$$

Find a quadratic equation having roots $\cos \frac{1}{5} \pi$ and $\cos \frac{3}{5} \pi$ and deduce the exact value of $\cos \frac{1}{5} \pi$. $\left(4x^2 - 2x - 1 = 0; \frac{1+\sqrt{5}}{4}\right) \quad [4]$

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- 33.** Use de Moivre's theorem to express $\cos^4 \theta$ in the form

$$a \cos 4\theta + b \cos 2\theta + c. \quad [4]$$

where a, b, c are constants to be found.

Hence evaluate

$$\int_0^{\frac{1}{4}\pi} \cos^4 \theta \, d\theta,$$

Leaving your answer in terms of π . $\left(a = \frac{1}{8}, b = \frac{1}{2}, c = \frac{3}{8}; \frac{1}{4} + \frac{3\pi}{32}\right)$ [3]

Oct/Nov 2011/13/5

34. Use de Moivre's theorem to prove that

$$\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}. \quad [6]$$

State the exact values of θ , between 0 and π , that satisfy $\tan 3\theta = 1$. $\left(\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{3\pi}{4}\right)$ [2]

Express each root of the equation $t^3 - 3t^2 - 3t + 1 = 0$ in the form $\tan(k\pi)$, where k is a positive rational number. $\left(t = \tan \frac{\pi}{12}, \tan \frac{5\pi}{12}, \tan \frac{3\pi}{4}\right)$ [3]

For each of these values of k , find the exact value of $\tan(k\pi)$. $\left(\tan \frac{\pi}{12} = 2 - \sqrt{3}; \tan \frac{5\pi}{12} = 2 + \sqrt{3}; \tan \frac{3\pi}{4} = -1\right)$ [3]

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35. By using the de Moivre's theorem to express $\sin 5\theta$ and $\cos 5\theta$ in term of $\sin \theta$ and $\cos \theta$, show that

$$\tan 5\theta = \frac{5t - 10t^3 + t^5}{1 - 10t^2 + 5t^4},$$

where $t = \tan \theta$. [5]

$$\begin{aligned} (\cos 5\theta &= \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta; \\ \sin 5\theta &= 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta) \end{aligned}$$

Show that the roots of the equation $x^4 - 10x^2 + 5 = 0$ are $\tan\left(\frac{1}{3}n\pi\right)$ for $n = 1, 2, 3, 4$. [2]

By considering the product of the roots of this equation, find the exact value of $\tan\left(\frac{1}{5}\pi\right) \tan\left(\frac{2}{5}\pi\right) \cdot (\sqrt{5})$ [3]

Oct/Nov 2010/1/10

36. (i) Write down the five fifth roots of unity. $\left(e^{\frac{2k\pi}{5}i}, k = 0, 1, 2, 3, 4\right)$ [2]
(ii) Hence find all the roots of the equation

$$z^5 + 16 + (16\sqrt{3})i = 0,$$

giving answers in the form $re^{iq\pi}$, where $r > 0$ and q is a rational number. Show these

roots on an Argand diagram. $\left(2e^{\left(-\frac{2\pi}{15} + \frac{2k\pi}{15}\right)i}\right)$ [4]

Let w be a root of the equation in part (ii),

(iii) Show that

$$\sum_{k=0}^4 \left(\frac{w}{2}\right)^k = \frac{3 + i\sqrt{3}}{2 - w}. \quad [3]$$

(iv) Identify the root for which $|2 - w|$ is least. $\left(w = 2e^{-\frac{2\pi}{15}i} \text{ or } 2e^{\frac{28\pi}{15}i}\right)$ [2]

May/June 2010/11/12/9

37. Use de Moivre's theorem to show that

$$\sin 5\theta = 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta. \quad [4]$$

Hence find all the roots of the equation

$$32x^5 - 40x^3 + 10x + 1 = 0$$

In the form $\sin(q\pi)$, where q is a positive rational number. [4]

$$\left(q = \frac{7}{30}, \frac{11}{30}, \frac{31}{30}, \frac{35}{30}, \frac{43}{30} \text{ or } q = \frac{7}{30}, \frac{19}{30}, \frac{31}{30}, \frac{43}{30}, \frac{55}{30} \text{ or } q = \frac{11}{30}, \frac{23}{30}, \frac{35}{30}, \frac{47}{30}, \frac{59}{30}\right)$$

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38. Use de Moivre's theorem to express $\sin^6 \theta$ in the form

$$a + b \cos 2\theta + c \cos 4\theta + d \cos 6\theta,$$

where a, b, c, d are constants to be found. $\left(a = \frac{5}{16}, b = -\frac{15}{32}, c = \frac{3}{16}, d = -\frac{1}{32}\right)$ [5]

Hence evaluate

$$\int_0^{\frac{1}{4}\pi} \sin^6 2x \, dx,$$

leaving your answer in terms of π . $\left(\frac{5}{64}\pi\right)$ [4]

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39. By considering $\sum_{k=0}^{n-1} (1 + i \tan \theta)^k$, show that

$$\sum_{k=0}^{n-1} \cos k\theta \sec^k \theta = \cot \theta \sin n\theta \sec^n \theta,$$

provided θ is not an integer multiple of $\frac{1}{2}\pi$. [7]

Hence or otherwise show that

$$\sum_{k=0}^{n-1} 2^k \cos\left(\frac{1}{3}k\pi\right) = \frac{2^n}{\sqrt{3}} \sin\left(\frac{1}{3}n\pi\right). \quad [2]$$

Given that $0 < x < 1$, show that

$$\sum_{k=0}^{n-1} \frac{\cos(k \cos^{-1} x)}{x^k} = \frac{\sin(n \cos^{-1} x)}{x^{n-1} \sqrt{1-x^2}}. \quad [4]$$

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40. Use de Moivre's theorem to express $\cos 8\theta$ as a polynomial in $\cos \theta$. (**$\cos 8\theta = 128 \cos^8 \theta - 256 \cos^6 \theta + 160 \cos^4 \theta - 32 \cos^2 \theta + 1$**) [5]

Hence

- (i) Express $\cos 8\theta$ as a polynomial in $\sin \theta$. (**$\cos 8\theta = 128 \sin^8 \theta - 256 \sin^6 \theta + 160 \sin^4 \theta - 32 \sin^2 \theta + 1$**) [2]
 (ii) Find the exact value of

$$4x^4 - 8x^3 + 5x^2 - x,$$

$$\text{where } x = \cos^2\left(\frac{1}{8}\pi\right). \left(-\frac{1}{16}\right) \quad [3]$$

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41. By considering $\sum_{n=1}^N z^{2n-1}$, where $z = e^{i\theta}$, show that

$$\sum_{n=1}^N \cos(2n-1)\theta = \frac{\sin(2N\theta)}{2 \sin \theta}, \quad [6]$$

where $\sin \theta \neq 0$.

Deduce that

$$\sum_{n=1}^N (2n-1) \sin\left[\frac{(2n-1)\pi}{N}\right] = -N \operatorname{cosec} \frac{\pi}{N}. \quad [4]$$

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42. Write down, in any form, all the roots of the equation $z^5 - 1 = 0$. (**$e^{\frac{2k\pi}{5}i}, k = 0, 1, 2, 3, 4$**) [2]

Hence find all the roots of the equation

$$(w-1)^4 + (w-1)^3 + (w-1)^2 + w = 0, \quad [4]$$

and deduce that none of them is real. $\left(1 + e^{\frac{2k\pi}{5}i}, k = 1, 2, 3, 4\right)$ [4]

Find the arguments of the two roots which have the smaller modulus. $\left(\pm \frac{2\pi}{5}\right)$
Oct/Nov 2007/1/9

43. (a) Use de Moivre's theorem to show that $\sin 8\theta$ can be expressed in the form

$$\sin \theta \cos \theta (a \sin^6 \theta + b \sin^4 \theta + c \sin^2 \theta + d),$$

where the value of the constant a is to be found and b, c, d are constants whose values need not be found. ($a = -128, b = 192, c = -80, d = 8$) [5]

(b) Use de Moivre's theorem to show that

$$\sum_{n=1}^N \frac{\sin n\theta}{2^n} = \frac{2^{N+1} \sin \theta + \sin N\theta - 2 \sin(N+1)\theta}{2^N(5 - 4 \cos \theta)}.$$
 [7]

May/June 2007/1/11

44. Prove de Moivre's theorem for a positive integral exponent:

$$\text{for all positive integer } n, (\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta. \quad [5]$$

Use de Moivre's theorem to show that

$$\cos 7\theta = 64 \cos^7 \theta - 112 \cos^5 \theta + 56 \cos^3 \theta - 7 \cos \theta. \quad [4]$$

Hence obtain the roots of the equation

$$128x^7 - 224x^5 + 112x^3 - 14x + 1 = 0$$

in the form $\cos q\pi$, where q is a rational number. $\left(q = \frac{2}{21}, \frac{8}{21}, \frac{14}{21}, \frac{20}{21}, \frac{26}{21}, \frac{32}{21}, \frac{38}{21}\right)$ [4]
Oct/Nov 2006/1/11

45. Given that $z = e^{i\theta}$ and n is a positive integer, show that

$$z^n + \frac{1}{z^n} = 2 \cos n\theta \quad \text{and} \quad z^n - \frac{1}{z^n} = 2i \sin n\theta \quad [2]$$

Hence express $\cos^7 \theta$ in the form

$$p \cos 7\theta + q \cos 5\theta + r \cos 3\theta + s \cos \theta,$$

where the constants p, q, r, s are to be determined. $\left(p = \frac{1}{64}, q = \frac{7}{64}, r = \frac{21}{64}, s = \frac{35}{64}\right)$ [4]

May/June 2006/1/9 (part)

46. Write down the fifth roots of unity. Hence, or otherwise, find all the roots of the equation

$$z^5 = -16 + (16\sqrt{3})i.$$

giving each root in the form $re^{i\theta}$. $\left(e^{\frac{2k\pi}{5}} \text{ where } k = 0, 1, 2, 3, 4; 2e^{\left(\frac{2\pi}{15} + \frac{2k\pi}{15}\right)i} \text{ where } k = 0, 1, 2, 3, 4\right)$ [4]

Oct/Nov 2005/1/1

47. Write down an expression in terms of z and N for the sum of the series

$$\sum_{n=1}^N 2^{-n} z^n. \quad [2]$$

$$\left(\frac{z}{2-z} \left[1 - \left(\frac{z}{2}\right)^N\right]\right)$$

Use de Moivre's theorem to deduce that

$$\sum_{n=1}^{10} 2^{-n} \sin\left(\frac{1}{10}n\pi\right) = \frac{1025 \sin\left(\frac{1}{10}\pi\right)}{2560 - 2048 \cos\left(\frac{1}{10}\pi\right)}. \quad [6]$$

Oct/Nov 2005/1/7

48. (i) Write down, in any form, all the complex roots of the equation

$$w^{12} = 1. \quad [2]$$

$$\left(e^{\frac{k\pi}{6}i}, \text{ where } k = 0, \pm 1, \pm 2, \pm 3, \pm 4, \pm 5, 6\right)$$

(ii) Explain why the equation

$$(z + 12)^{12} = z^{12} \quad (*)$$

has exactly 10 non-real roots and show that they may be expressed in the form

$$-1 - i \cot\left(\frac{1}{12}k\pi\right).$$

where $k = \pm 1, \pm 2, \pm 3, \pm 4, \pm 5$. [6]

(because it's a polynomial equation of degree 11 with real coefficients and has one real root -1)

(iii) Show that

$$\left\{-1 - i \cot\left(\frac{1}{12}k\pi\right)\right\} \left\{-1 + i \cot\left(\frac{1}{12}k\pi\right)\right\} = \operatorname{cosec}^2\left(\frac{1}{12}k\pi\right). \quad [1]$$

(iv) Given that the product of the roots of (*) is $-\frac{512}{3}$, find the value of

$$\sin^2\left(\frac{1}{12}\pi\right)\sin^2\left(\frac{2}{12}\pi\right)\sin^2\left(\frac{3}{12}\pi\right)\sin^2\left(\frac{4}{12}\pi\right)\sin^2\left(\frac{5}{12}\pi\right). \quad [2]$$

$$\left(\frac{3}{512}\right)$$

May/June 2005/1/10

49. Write down all the 8th roots of unity. $\left(e^{\frac{k\pi}{4}i}, \text{where } k = 0, \pm 1, \pm 2, \pm 3, 4\right)$ [2]

Verify that

$$(z - e^{i\theta})(z - e^{-i\theta}) \equiv z^2 - (2 \cos \theta)z + 1. \quad [1]$$

Hence express $z^8 - 1$ as the product of two linear factors and three quadratic factors, where all coefficients are real and expressed in a non-trigonometric form. $((z - 1)(z + 1)(z^2 + 1)(z^2 - \sqrt{2}z + 1)(z^2 + \sqrt{2}z + 1))$ [5]

Oct/Nov 2004/1/6

50. Find the roots of the equation

$$z^3 = -(4\sqrt{3}) + 4i,$$

giving your answers in the form $re^{i\theta}$, where $r > 0$ and $0 \leq \theta < 2\pi$. $\left(2e^{\frac{5\pi}{18}i}, 2e^{\frac{17\pi}{18}i}, 2e^{\frac{29\pi}{18}i}\right)$ [5]

Denoting these roots by z_1, z_2, z_3 , show that, for every positive integer k ,

$$z_1^{3k} + z_2^{3k} + z_3^{3k} = 3 \left(2^{3k} e^{\frac{5}{6}k\pi i}\right). \quad [4]$$

May/June 2004/1/7

51. Given that $z = e^{i\theta}$ and n is a positive integer, show that

$$z^n + \frac{1}{z^n} = 2 \cos n\theta \quad \text{and} \quad z^n - \frac{1}{z^n} = 2i \sin n\theta \quad [2]$$

Hence express $\sin^6 \theta$ in the form

$$p \cos 6\theta + q \cos 4\theta + r \cos 2\theta + s \cos \theta,$$

where the constants p, q, r, s are to be determined. $\left(p = -\frac{1}{32}, q = \frac{3}{16}, r = -\frac{15}{32}, s = \frac{5}{16}\right)$ [4]

Oct/Nov 2003/1/8 (part)

52. Use de Moivre's theorem to show that

$$\cos 6\theta = 32 \cos^6 \theta - 48 \cos^4 \theta + 18 \cos^2 \theta - 1. \quad [5]$$

Hence solve the equation

$$64x^6 - 96x^4 + 36x^2 - 1 = 0,$$

Giving each root in the form $\cos k\pi$. $\left(k = \frac{1}{9}, \frac{4}{9}, \frac{7}{9}, \frac{10}{9}, \frac{13}{9}, \frac{16}{9}\right)$ [4]

May/June 2003/1/6

53. Given that $z = \cos \theta + i \sin \theta$, show that

(i) $z - \frac{1}{z} = 2i \sin \theta,$ [1]

(ii) $z^n + z^{-n} = 2 \cos n\theta.$ [2]

Hence show that

$$\sin^6 \theta = \frac{1}{32} (10 - 15 \cos 2\theta + 6 \cos 4\theta - \cos 6\theta). \quad [3]$$

Find a similar expression for $\cos^6 \theta$, and hence express $\cos^6 \theta - \sin^6 \theta$ in the form $a \cos 2\theta + b \cos 6\theta$. $(\cos^6 \theta = \frac{1}{32} (\cos 6\theta + 6 \cos 4\theta + 15 \cos 2\theta + 10), a = \frac{15}{16}, b = \frac{1}{16})$ [3]

Oct/Nov 2002/1/7

54. Given that $w_n = 3^{-n} \cos 2n\theta$ for $n = 1, 2, 3, \dots$, use de Moivre's theorem to show that

$$1 + w_1 + w_2 + w_3 + \dots + w_{N-1} = \frac{9 - 3 \cos 2\theta + 3^{-N+1} \cos 2(N-1)\theta - 3^{-N+2} \cos 2N\theta}{10 - 6 \cos 2\theta}. \quad [7]$$

Hence show that the infinite series

$$1 + w_1 + w_2 + w_3 + \dots$$

is convergent for all values of θ , and find the sum to infinity. $\left(\frac{9-3 \cos 2\theta}{10-6 \cos 2\theta}\right)$ [2]

May/June 2002/1/9